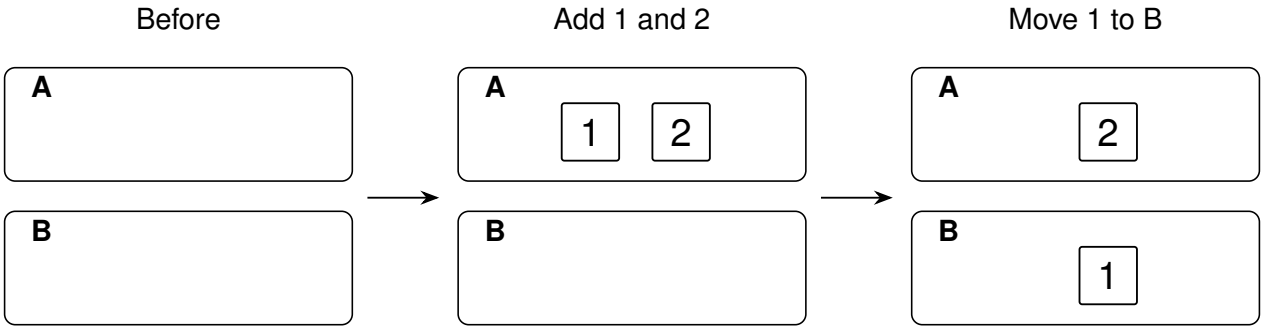
Both bowls start empty. Each day, put the next two numbered cards in A, then move A's smallest card to B. Cards in B stay there.



Problem 1: Run six days with cards 1–12. Can the bowls ever end a day with different numbers of cards? You may continue to twelve days with cards 1–24.

Problem 2: Restart. Find rules for the day any numbered card enters A and the day it moves to B. Test your rules on cards 3, 6, and 10, and predict both days for card 20 without running twenty days.

3	6	10	20
Enters day _____	Enters day _____	Enters day _____	Enters day _____
Moves day _____	Moves day _____	Moves day _____	Moves day _____

Problem 3: Start again. Each day, move the larger of today's two new cards to B, instead of A's smallest card. Could somebody tell which rule you used from the bowl counts alone? Find a way to show the difference using the cards.

Problem 4: Start again, but now you choose which card in A moves to B each day. After six days, can B hold the six cards in each row?

1	3	5	7	9	11

2	3	5	7	8	11

1	4	6	9	11	12

Problem 5: Imagine the numbered cards continuing forever, with two new cards each day. Call a card settled in a bowl if, after some day, it stays in that bowl forever. For each rule, decide where all the cards settle. Can that arrangement happen on a finite day?

Move A's smallest card

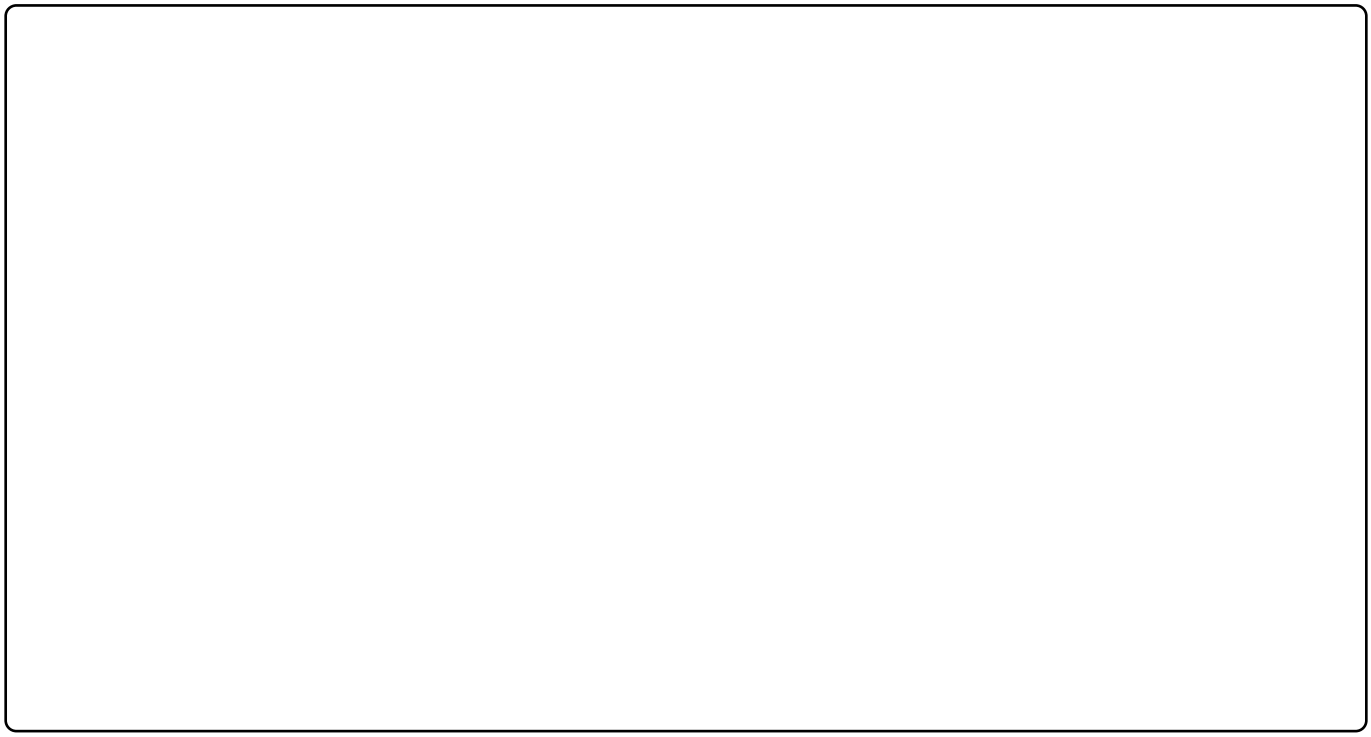
A	B

Move the larger new card

A	B

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16
17	18	19	20
21	22	23	24

A



B